

Healthcare Quality Automation & Feedback System

Lean Six Sigma Yellow Belt Case Study

Process Improvement | Excel/VBA | Quickbase

Project Snapshot

Project Duration

July 2025 - November 2025

Automated Tool Implementation

September 2025

Team

1 Project Leader, 2 QAs

Domain

Healthcare Quality Operations

Certification

Lean Six Sigma Yellow Belt - December 2025

Project at a Glance

A healthcare quality audit and feedback process was improved by transitioning from a manual, multi-file Excel-based workflow to a centralized and automated Quickbase solution.

The improvement journey built upon an Excel/VBA automation solution developed several months before the Yellow Belt project, which was later scaled through Quickbase to improve efficiency, accuracy, feedback management, reporting, and visibility.

Key Results

42.50 → 28.42 min

Average Audit Time

33.13%

Efficiency Improvement

1,000+ / month

Audit Volume

97%

Positive User Response

~\$9,500

Reported Annual Savings

Portfolio Version

Generalized for public sharing. Client names, proprietary screenshots, internal datasets, and confidential process documentation are excluded.

1. Business Problem & Current State

The quality audit process involved reviewing production transactions against defined audit parameters, identifying errors, calculating scores, providing feedback, and preparing performance reports.

The existing workflow relied heavily on **individual Excel files and manual activities**, creating opportunities for errors, delays, duplicate audits, and limited visibility.

Key Challenges

- **Multiple Excel files** - Each QA maintained individual audit records that required later consolidation.
- **Manual data entry** - Case details, error parameters, comments, and scores involved repetitive manual activities.
- **Manual feedback** - Feedback emails were prepared and sent manually, creating a risk of missed or incorrect information.
- **Duplicate audits** - Independently selected cases created a possibility of duplicate audits.
- **Manual reporting** - Daily, weekly, and monthly reports required manual consolidation and analysis.
- **Limited visibility** - Leaders and operations did not always have real-time access to the latest audit information.

Current-State Workflow

Before Improvement

Sampling → Case Selection → Audit → QA File → Scoring → Feedback → Consolidation → Reporting

Baseline

Measure	Baseline
Monthly audits	~500
QA team	3 QAs
Average audit time	42.50 min
Feedback	Manual
Reporting	Manual consolidation

Improvement Opportunity

The process contained repetitive manual activities across **audit execution, data entry, feedback, consolidation, and reporting**.

The key opportunity was to **centralize and automate the workflow**, reducing manual effort while improving accuracy, feedback tracking, reporting visibility, and audit capacity.

2. Lean Six Sigma Analysis

SIPOC Analysis

SIPOC was used to establish a high-level view of the audit and feedback process and understand the relationship between the key stakeholders, inputs, process activities, outputs, and customers.

Suppliers	Inputs	Process	Outputs	Customers
QCA Team	Audit parameters	Collect audit data	Audit scores	QCA Team
Process Owners	Error logs	Evaluate transactions	Error trend reports	Team Leads
Internal Systems	QA guidelines	Identify & analyze errors	Individual feedback	Operations
	Scoring criteria	Calculate audit scores	Performance dashboards	Agents
		Provide feedback & reports		

FMEA - Risk Identification

FMEA was used to identify potential failure modes within the existing audit and feedback process and assess where process improvements could have the greatest impact.

Key Risk Areas Identified

Process Risk	Improvement Opportunity
Manual data entry	Standardize and automate data capture
Duplicate case audits	Introduce system-level duplicate prevention
Incorrect feedback details	Automate data retrieval and feedback generation
Missed feedback	Introduce automated feedback and tracking
Manual error recording	Use structured error capture
Reporting delays	Centralize audit data and reporting
Limited visibility	Provide dashboards and filterable information

Analysis Outcome

The analysis highlighted that many of the process risks originated from **manual data handling and dependency on multiple files**.

The improvement therefore focused on three key areas:

1. Centralization

Create a single source of audit information rather than relying on multiple files.

2. Automation

Reduce repetitive manual activities in scoring, feedback, and data handling.

3. Visibility

Enable real-time access to audit information, error trends, feedback status, and performance data.

3. Improvement Journey & Solution

Stage 1 — Excel/VBA Automation

Several months before the Yellow Belt project, I developed an **Excel/VBA-based automation solution** to address recurring manual activities in the audit and feedback process.

The tool was designed to support my individual audit workflow rather than act as a centralized audit database.

Key Capabilities

- Retrieved agent details using **VLOOKUP/VBA** based on the case or agent information entered.
- Pulled the audit score from the personal audit-tracking workbook.
- Reduced repetitive manual data entry and copy-paste activities.
- Generated the feedback email template through a **VBA-triggered email function**.
- Helped reduce errors while preparing feedback communication.

Limitation

Although the Excel/VBA solution reduced manual effort, the process still depended on **individual files, manual consolidation, and separate tracking**.

This created a need for a more scalable and centralized solution.

Stage 2 — Quickbase Migration

The Yellow Belt project focused on migrating the process into **Quickbase** to centralize the audit workflow and address the limitations of the multi-file process.

Why Quickbase?

The centralized platform enabled the process to move from multiple individual files to a **single source of audit information**.

It eliminated the need to maintain separate files for different QAs, audit periods, and reporting requirements.

Users could retrieve required information through **filters and dashboards**, while audit data became available with real-time visibility.

Before → After

Before

Sampling → Case Selection → Audit → QA File → Manual Scoring → Feedback → Consolidation → Reporting

After

Bulk Upload → Centralized Audit → Error Capture → Auto Score → Auto Feedback → Tracking → Dashboard

Centralized Quickbase Solution

The Quickbase solution enabled:

Capability	Improvement
Case Upload	QAs can bulk upload cases for auditing
Duplicate Control	Duplicate cases are prevented
Audit Capture	Errors and comments are recorded in structured fields
Scoring	Final audit score is calculated automatically
Agent Details	Information is retrieved from the roster
Feedback	Feedback email is triggered from the tool
Tracking	Feedback activity can be tracked
Dashboards	Error trends and performance are visible
Filtering	Required audit information can be filtered
Visibility	Centralized, real-time information is available

My Role in the Migration

I proposed the migration to Quickbase and worked with the developer during implementation. I tested the solution, identified required changes, provided feedback during development, and supported the transition after implementation.

Following successful implementation, I **trained fellow QAs and conducted a walkthrough for the Operations team** to support adoption.

4. Results & Impact

Measurable Improvement

The implementation of the centralized audit and feedback solution resulted in improvements in audit efficiency, capacity, feedback management, and process visibility.

Key Performance Results

Metric	Before	After	Improvement
Average Audit Time	42.50 min	28.42 min	14.08 min reduction
Efficiency	Baseline	33.13%	Improved efficiency
Monthly Audit Volume	~500	1,000+	Increased capacity
QA Team	3 QAs	3 QAs	Higher capacity with same team
Annual Savings	-	~\$9,500	Reported project impact
User Response	-	97% positive	Strong acceptance

Process Improvements

Before → After

Manual Audit Activities	→	Automated & Structured Workflow
Multiple Files	→	Centralized Audit Data
Manual Scoring	→	Automatic Score Calculation
Manual Feedback	→	Automated Feedback & Tracking
Manual Reporting	→	Dashboards & Filterable Reports
Limited Visibility	→	Real-Time Process Visibility

Business & Operational Impact

Higher Audit Capacity

Monthly audit capacity increased from approximately **500 to 1,000+ audits** with the same team of 3 QAs.

Reduced Audit Time

Average audit time reduced from **42.50 to 28.42 minutes**, representing a **33.13% efficiency improvement**.

Improved Feedback Management

Automated feedback triggering reduced manual email preparation and established feedback tracking.

Improved Data & Reporting

Centralized audit data enabled easier filtering, trend analysis, dashboards, and real-time visibility.

Reported Financial Impact

Approximately **\$9,500 in annual savings** was reported as part of the project impact.

5. Project Takeaways & Skills Demonstrated

Project Timeline

Milestone	Timeline
Initial Excel/VBA automation developed	Several months before YB project
Improvement opportunity identified & YB project initiated	July 2025
Quickbase development & testing	August–September 2025
Quickbase implementation	September 2025
Stabilization, training & adoption	September–November 2025
Project completed	November 2025
Lean Six Sigma Yellow Belt certification	December 2025

Key Learning

The project demonstrated how an existing Excel/VBA automation could be **scaled into a centralized workflow** through Quickbase.

The improvement combined **process analysis, risk identification, automation, standardization, and user adoption** to improve audit efficiency, capacity, feedback management, reporting, and visibility.

Skills Demonstrated

Process Improvement

SIPOC • FMEA • Workflow Analysis • Standardization

Automation & Technology

Excel • VBA • VLOOKUP • Quickbase • Dashboards

Quality & Data Analysis

Quality Auditing • Error Analysis • Scoring • Trend Reporting

Stakeholder & Change Management

Developer Collaboration • Testing • Training • Operations Walkthrough

Certification

Lean Six Sigma Yellow Belt - December 2025

Project Outcome

From a manual, multi-file audit process to a centralized and automated quality workflow.